NeST decode as a multi-branch tree: selection (beam) and refinement

each node is a kept SE(3) cell (rotation HEALPix tile × translation octree octant); edges are the child conditionals $p(c \mid \text{parent})$; gold = the beam-MAP path

root: 72-way rotation ($g = 0$)

$\ell = 0$

nside 1
$n_{roll} = 6$
$N = 72 \cdot 8^0$

refine ×8

0.57 0.31 0.05

64-way split, keep top-3 (beam)

$\ell = 1$

nside 2
$n_{roll} = 12$
$N = 72 \cdot 8^1$

0.58 0.28 0.43

$\ell = 2$

nside 4
$n_{roll} = 24$
$N = 72 \cdot 8^2$

0.20 0.18 0.35

$\ell = 3$

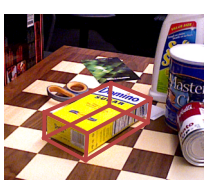
nside 8
$n_{roll} = 48$
$N = 72 \cdot 8^3$

0.20 0.32 0.19

$\ell = 4$

nside 16
$n_{roll} = 96$
$N = 72 \cdot 8^4$

⋮ **refine to $\ell = 5$ leaf**



beam-MAP leaf → 6D pose (R, t)

node = HEALPix tile (rotation, hue=roll)
+ octree octant (translation)
cell refines ×4 (tile), ×8 (octant) / level

beam-MAP path

other kept beam branches

faint stubs = 61 pruned children / node